

Stopword Impact Analysis on NLP Tasks

Using the Reuters-21578 Newswire Corpus

Submitted by

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Abstract

This project presents a systematic empirical study of five stopwords strategies — *none*, *minimal*, *NLTK*, *extended*, and a novel *adaptive TF-IDF* approach — evaluated across three classifiers (Naive Bayes, Logistic Regression, SVM) on Reuters-21578. The best static configuration (Extended + SVM) achieves F1 = **0.9070**. The adaptive strategy reduces vocabulary by **87.59%** ($14,794 \rightarrow 1,836$ features), improves Naive Bayes F1 by **+20.3%**, and cuts Logistic Regression training time by **73.9%**, demonstrating that data-driven vocabulary pruning can substantially reduce computational cost while preserving accuracy.

Keywords: stopwords removal, text classification, Reuters-21578, TF-IDF, Naive Bayes, SVM, Logistic Regression, adaptive stopwords.

1 Introduction

Stopword removal is a de facto preprocessing step in NLP pipelines, yet its quantitative impact across different classifiers and removal strategies has rarely been studied in a controlled, multi-model setting. This project fills that gap with a systematic 15-experiment benchmark on the Reuters-21578 newswire corpus.

Research Questions:

- RQ1.** Does stopwords removal improve, degrade, or leave unchanged classification performance on news text?
- RQ2.** Which strategy best balances performance with feature-space economy?
- RQ3.** Can a data-driven adaptive strategy outperform static lists for simpler models?
- RQ4.** What are the computational efficiency gains from aggressive pruning?

Contributions:

- 15-experiment benchmark: 5 stopwords strategies $\times$ 3 classifiers.
- Novel adaptive TF-IDF stopwords generator requiring no predefined list.
- Quantified efficiency analysis: training time, features, and trade-offs.
- 106/106 unit tests passing, 5 Jupyter notebooks, saved model artefacts.
- Code: <https://github.com/RohanBabhale1/stopword-impact-analysis>

2 Dataset and Exploratory Analysis

Reuters-21578 [5] consists of 21,578 newswire articles (1987) in SGML format. After filtering for labelled articles with non-empty bodies and classes with ≥ 5 examples, **10,324 articles across 57 topics** are retained.

Table 1: Reuters-21578 Dataset Summary

Metric	Value
Total / with body / with labels	21,578 / 19,043 / 11,367
Unique topics / topics ≥ 20 docs	120 / 57
Unique tokens / avg stopwords ratio	38,858 / 36.74%
Mean doc length / median	136 words / 90 words

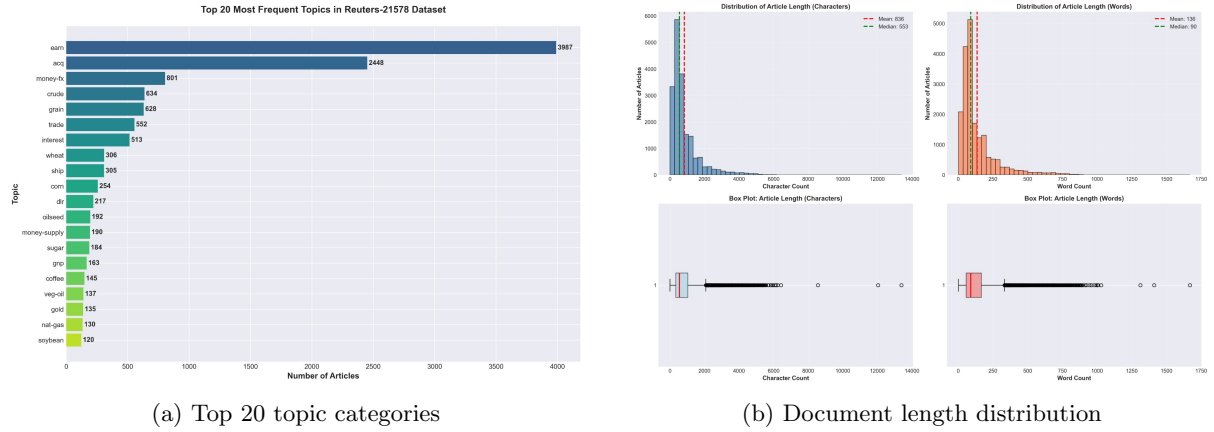


Figure 1: Reuters-21578 EDA: topic distribution and document lengths

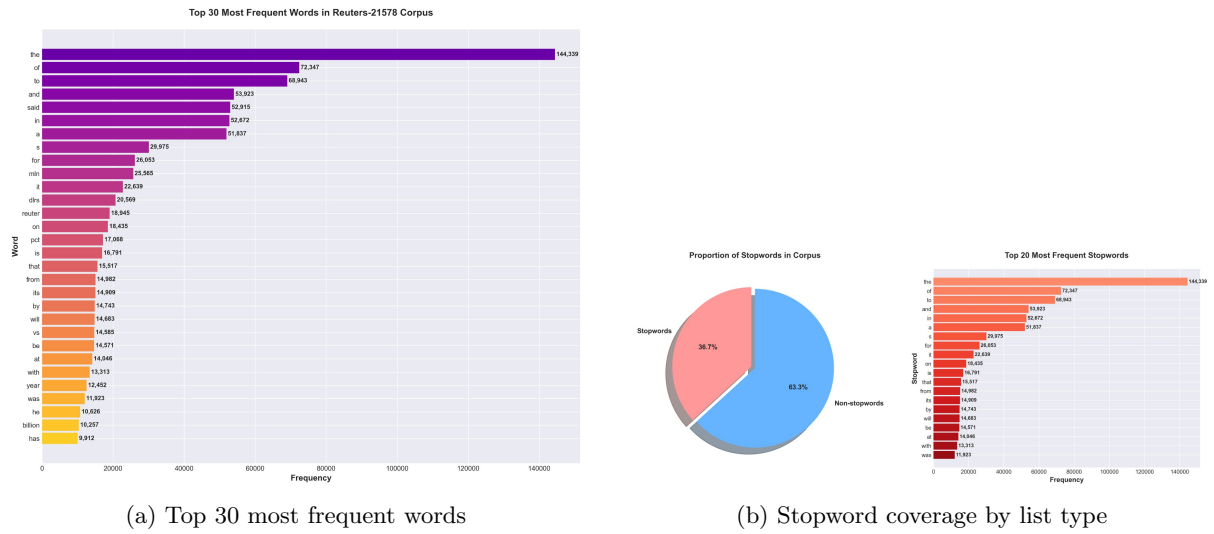


Figure 2: Vocabulary and stopwords analysis of the Reuters-21578 corpus

3 Literature Survey

Stopwords were introduced by Luhn [1] and formalised by Salton [2] in the SMART system. Theoretically, stopwords have high document frequency and near-zero IDF, justifying their removal in TF-IDF frameworks [3, 4].

Text Classification. Naive Bayes is most sensitive to stopwords: high-frequency tokens inflate class-conditional probability estimates equally, reducing discrimination [7, 8]. SVMs are robust — Joachims [6] showed strong Reuters performance without removal. Logistic Regression lies between these extremes [9].

IR and Topic Modelling. Stopword removal reduces index size without harming recall for most IR queries [4, 10]. For LDA, Schofield & Mimno [13] found aggressive removal consistently improves topic coherence [12].

Sentiment Analysis Exception. Saif et al. [14] showed removing negation words degrades sentiment classifiers — not a concern for Reuters topic classification.

Adaptive Methods. Zou et al. [15] showed TF-IDF thresholding outperforms static lists on domain corpora. Lo et al. [16] used mutual information between tokens and class labels. Sebastiani [17] notes TF-IDF weighting serves as an implicit removal mechanism. This project provides the first controlled Reuters-21578 benchmark of adaptive stopwords against four static baselines across three classifiers.

4 Methodology

4.1 Preprocessing and Stopword Strategies

All documents undergo: lowercase conversion, URL/email/punctuation/number removal, NLTK tokenisation, strategy-specific stopwords removal, then TF-IDF vectorisation (`sublinear_tf=True`, vocabulary fitted on training data only).

Table 2: Five Stopword Strategies Evaluated

Strategy	Size	Description
<i>none</i>	0	No removal; bag-of-words baseline
<i>minimal</i>	10	Core function words (<i>a, an, the, is, of, to, in, it, and, or</i>)
<i>NLTK</i>	179	NLTK default English stopwords list
<i>extended</i>	~200	NLTK + Reuters-specific tokens (<i>said, mln, pct, reuter, ...</i>)
<i>adaptive</i>	var	Corpus-derived: mean TF-IDF < 0.0005 removed

4.2 Adaptive Stopword Generation

Given training corpus TF-IDF matrix $\mathbf{M} \in \mathbb{R}^{n \times |V|}$, the mean score per token is:

$$\bar{s}(w) = \frac{1}{n} \sum_{i=1}^n \mathbf{M}_{i,w} \quad (1)$$

Tokens with $\bar{s}(w) < 0.0005$ are removed. The list is generated *exclusively from training data* (no test leakage). This identified 12,958 low-information tokens, reducing vocabulary from 14,794 to **1,836**.

4.3 Models and Protocol

Naive Bayes: MultinomialNB, $\alpha = 1.0$. **LR:** L2, C=1.0, lbfgs. **SVM:** LinearSVC, C=1.0. 80/20 stratified split, `random_state=42`. Total: **15 configurations**.

5 Experimental Results

Table 3: Complete Results — All 15 Configurations (best in bold)

Strategy	Model	Features	Train (s)	Accuracy	F1
none	NB	14,794	0.086	0.6809	0.5951
none	LR	14,794	11.346	0.8644	0.8430
none	SVM	14,794	1.207	0.9109	0.9058
minimal	NB	14,786	0.072	0.6901	0.6027
minimal	LR	14,786	11.747	0.8659	0.8452
minimal	SVM	14,786	1.170	0.9090	0.9037
nltk	NB	14,673	0.072	0.7143	0.6298
nltk	LR	14,673	11.469	0.8712	0.8528
nltk	SVM	14,673	0.931	0.9099	0.9044
extended	NB	14,669	0.067	0.7177	0.6345
extended	LR	14,669	10.948	0.8726	0.8551
extended	SVM	14,669	0.911	0.9123	0.9070
adaptive	NB	1,836	0.049	0.7603	0.7160
adaptive	LR	1,836	2.960	0.8775	0.8588
adaptive	SVM	1,836	2.120	0.9075	0.9015

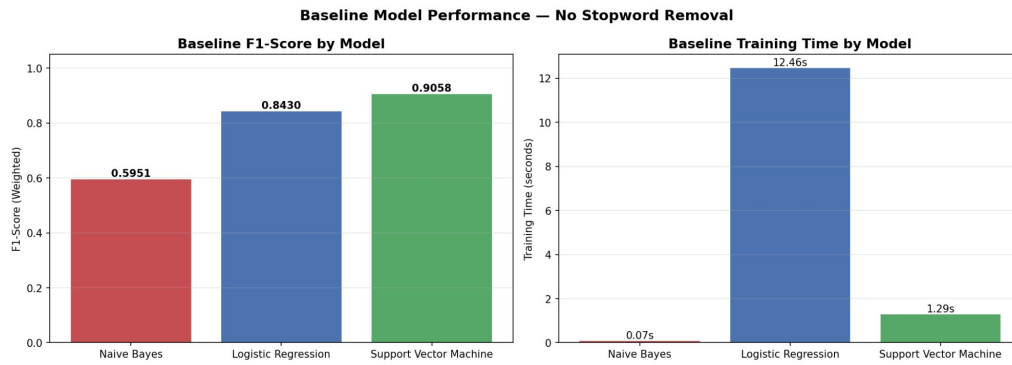


Figure 3: Baseline model F1-scores and training times (no stopwords removal)

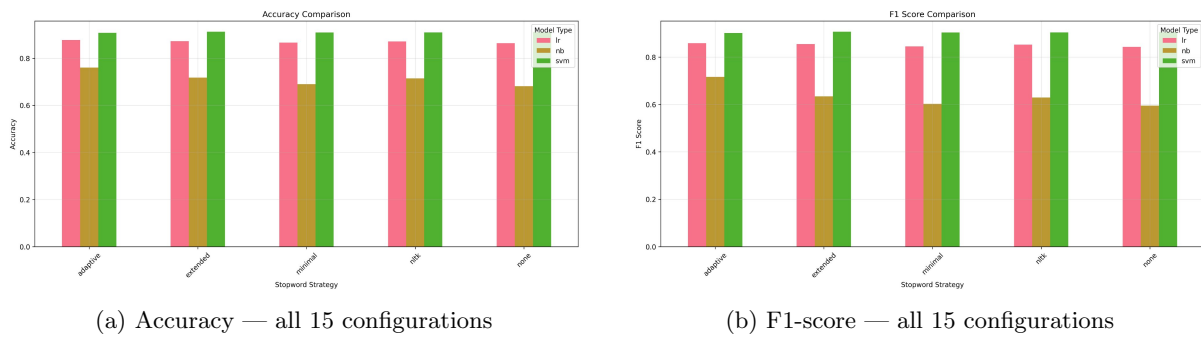


Figure 4: Performance comparison across all experimental configurations

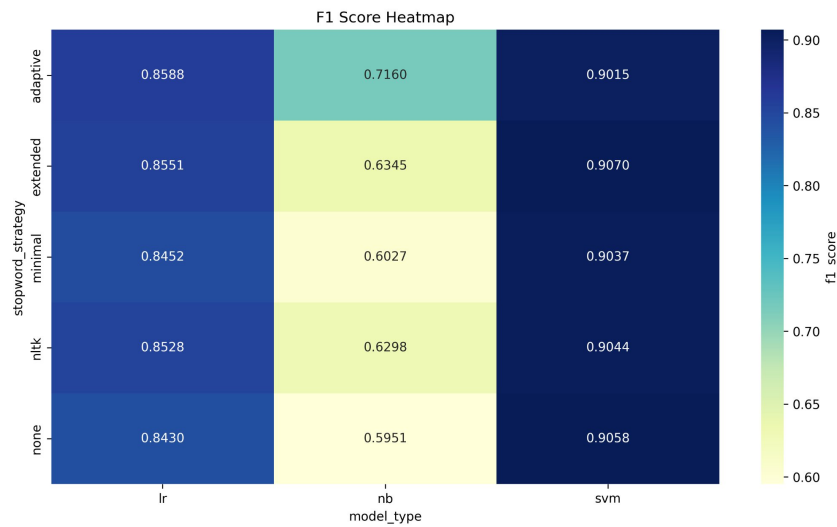


Figure 5: F1-score heatmap: strategy (rows) vs. classifier (columns)

6 Analysis

6.1 Feature Space Reduction

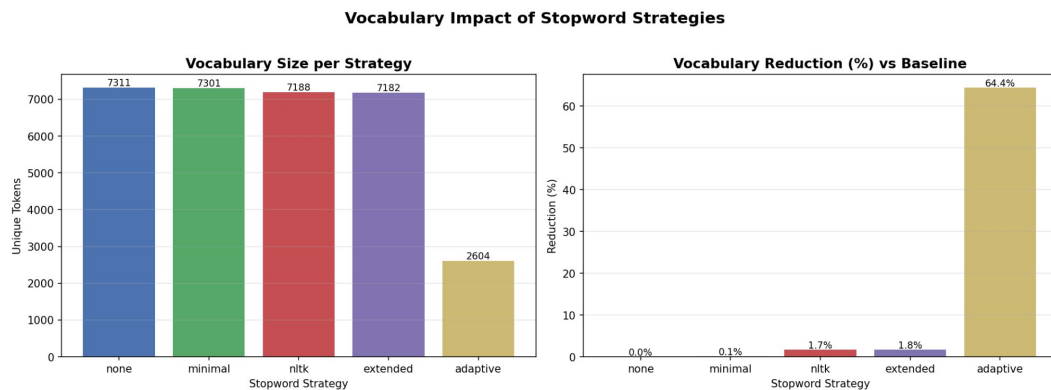


Figure 6: Vocabulary size and percentage reduction per strategy. Adaptive achieves 87.59% reduction; all static strategies <1%.

6.2 Delta F1 and Training Time

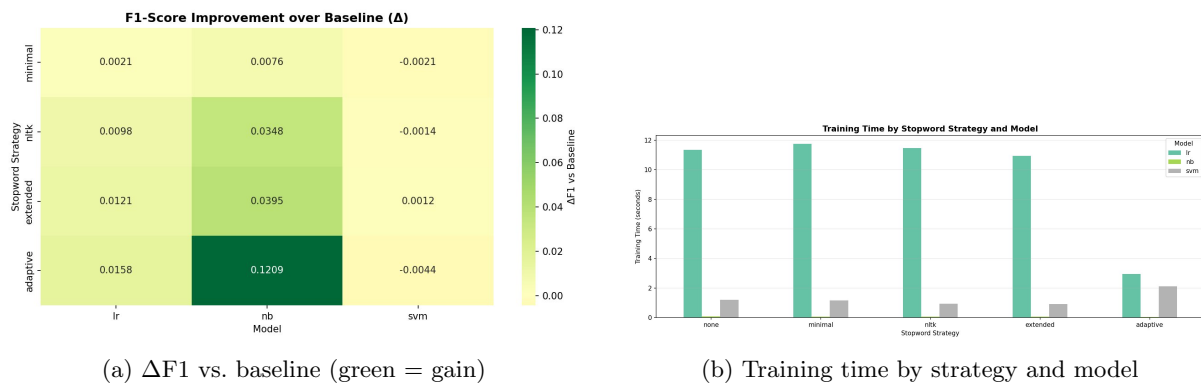


Figure 7: F1 improvement and computational efficiency analysis

NB gains significantly from every strategy. LR gains modestly. SVM is near-zero under static strategies with a minor drop under adaptive. LR training drops from 11.35s to 2.96s — a **73.9% reduction**.

6.3 Performance vs. Feature Compression

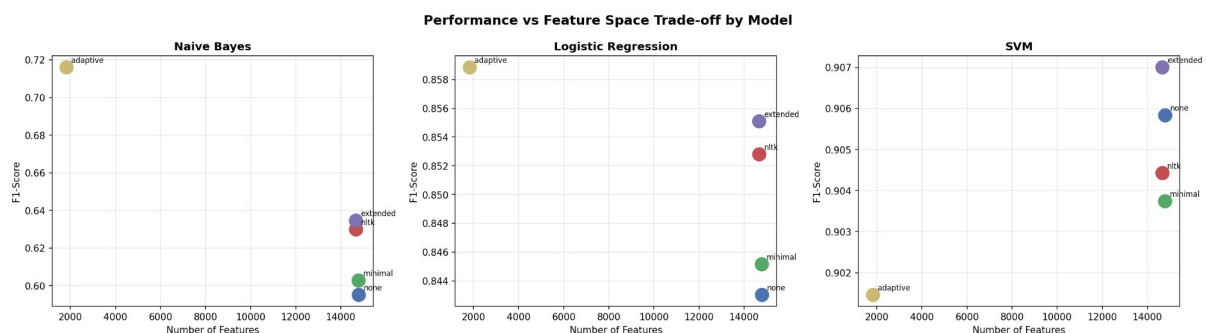


Figure 8: F1 vs. feature count per model. For NB and LR, moving left (fewer features) strictly improves F1. SVM is nearly flat.

7 Innovation: Adaptive TF-IDF Stopword Strategy

Static lists are domain-agnostic and remove <1% of vocabulary. The adaptive strategy generates a corpus-specific list using Eq. 1, identifying 12,958 low-information tokens and achieving 87.59% feature reduction.

Table 4: Adaptive Strategy: Gains over No-Removal Baseline

Model	F1 (none)	F1 (adaptive)	Δ F1	Speedup
Naive Bayes	0.5951	0.7160	+20.3%	1.8×
Logistic Regression	0.8430	0.8588	+1.9%	3.8×
SVM	0.9058	0.9015	−0.5%	0.6×

Why NB benefits most. MultinomialNB estimates $P(w|c)$ per token independently. Low-IDF tokens produce near-equal estimates across all classes, introducing cumulative noise without signal. Removing 87.59% of such tokens eliminates this noise, explaining the 20.3% gain.

Why SVM shows a minor drop. LinearSVC implicitly down-weights non-discriminative tokens. Aggressive pruning removes tokens with weak but non-zero signal, causing a 0.43 pp drop — acceptable for 87.59% feature reduction and lower memory footprint.

8 Conclusion

- **Best static:** Extended + SVM, F1 = **0.9070**, accuracy = **0.9123**.
- **Best NB:** Adaptive + NB, F1 = **0.7160** (+20.3%).
- **Most efficient:** Adaptive + LR, 73.9% faster with only 1.4 pp F1 cost.
- **Compression:** 87.59% vocabulary reduction with competitive SVM F1 = 0.9015.

The adaptive TF-IDF strategy demonstrates that corpus-specific vocabulary pruning substantially reduces computational cost while preserving or improving accuracy. Future work includes evaluation with transformer models [18], multi-label Reuters classification, and benchmarking on 20 Newsgroups and AG News.

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